TEMPLE COLLEGE
WORKFORCE DEVELOPMENT

IN COLLABORATION WITH CHIEFOS

WEEK PREVIEW · LEARNING OBJECTIVES

WEEK 6 · DOCKER + PHASE 1 GATE

MODULE 02 · IMAGES, SIDEBAR & CHECKPOINT

We skipped last week's Zoom — so tonight recaps Week 5 (install + first containers), then images vs containers, the Docker sidebar, and a walkthrough of the Phase 1 gate.

MON AUG 24 + WED AUG 26 · 6:00–8:00 PM · ZOOM · LABS 02:00–06

This week closes Phase 1. Week 5 was Docker Engine, `hello-world`, and nginx on port 8080 (02:00–03). Tonight we catch that up live, then teach images vs containers and the VS Code Docker sidebar (02:04–05). Wednesday is the clinic plus the timed, no-help **Module 02 / Phase 1 gate** (02:06). Labs — not attendance — gate progress.

TONIGHT'S LEARNING OBJECTIVES

02:01
INSTALL ENGINE

Docker Engine from the official apt repo — not Desktop. Join the `docker` group; `newgrp` or re-login.

02:02–03
FIRST CONTAINERS

`docker run hello-world`, then nginx with `-d -p 8080:80`. Host 8080 → container 80.

02:04
IMAGE VS CONTAINER

Recipe vs meal. `docker ps` vs `ps -a`. `rm` removes a container; `rmi` removes an image.

02:05
DOCKER SIDEBAR

Whale icon finally populated. Stop / remove / logs from the UI — same as the CLI.

WEEK 5 RECAP (SKIPPED ZOOM)

- **02:00** Setup check — Linux, sudo, notes file, Hub reachability
- **02:01** Install Engine from Docker's apt repo; add yourself to `docker`
- **02:02** First container — "Unable to find image locally" is **not** an error
- **02:03** nginx welcome page at `localhost:8080` (or Ports tab if SSH)
- Uplink hunt (Week 4) was the network dress rehearsal this install rides on

KNOW BEFORE YOU TYPE

- We want **Docker Engine** (Linux daemon), not Docker Desktop
- `permission denied … docker.sock` → log out / `newgrp docker`
- WSL: if `systemctl` fails, `sudo service docker start`
- Forgot `-d`? Terminal "hangs" — nginx is in the foreground. **Ctrl+C**, re-run
- Wednesday's gate = from memory, no copy-paste, no neighbor help

MONDAY · INSTRUCTION (WEEK 5 CATCH-UP + WEEK 6)

CLOCK	BLOCK	WHAT WE DO LIVE	LAB IDS
6:00	Open + recap	Sign-in · Week 5 catch-up poll: who can `docker run`? · what Docker is (Engine, not Desktop)	—
6:20	Week 5 live	Install Engine if needed · `hello-world` · nginx `-d -p 8080:80` · browser / Ports / `curl`	02:00–03
7:00	Images vs containers	Recipe vs meal · `docker images` · `ps` vs `ps -a` · `rm` vs `rmi` · cleanup	02:04
7:30	Sidebar + gate walk	Whale icon populated · stop/remove from UI · walk **02:06** shape (do not sit the gate tonight)	02:05–06

WEDNESDAY · LAB CLINIC + PHASE 1 GATE

CLOCK	BLOCK	WHAT WE DO LIVE	LAB IDS
6:00	Clinic triage	Install leftovers · docker-group / `newgrp` · WSL `service docker start` · port-forward paths	02:00–05
6:40	Phase 1 gate	Timed, no-help run of **02:06** — nginx named `final` on 8080, curl, sidebar screenshot	02:06
7:30	Redo + catch-up	Miss a section? Re-open that lesson. Anyone still on install gets 1:1 help	02:06
7:50	Close	Gate screenshot in D2L · preview Phase 2 (FastAPI)	—

If time is short tonight: protect this order — Engine actually runs → nginx on 8080 → image vs container → gate walkthrough. Push the sidebar (02:05) and leftover install triage to Wednesday if install eats the first hour.

W5	02:04	02:06	P2
RECAP TONIGHT	IMAGE VS CONTAINER	PHASE 1 GATE	UNLOCKS FASTAPI

LEARNING OBJECTIVES — WHAT "DONE" LOOKS LIKE

Each row is one D2L learning object. Orange tags = Week 5 catch-up we do live tonight. Do them **in order** — the Phase 1 gate assumes Engine, hello-world, nginx, and image vs container are warm.

CORE MODULE 02 · DOCKER

Mon catch-up + instruction · Wed gate

#	LAB	LEARNING OBJECTIVE	PASS / SCREENSHOT
02:00	Setup Check	W5 Confirm Linux, sudo, `curl` / `gpg`, and Module 01 notes before install.	Paste checkpoint: `sudo whoami && which curl gpg && ls ~/docker-course/notes/notes.txt`
02:01	Install Docker Engine	W5 Install Engine from Docker's apt repo; add yourself to the `docker` group.	`docker --version` · `docker info` · `groups` shows docker · notes line
02:02	First Container	W5 Run `docker run hello-world` and explain pull → run → exit (pull line is not an error).	`docker images | grep hello-world` · `docker ps -a | grep hello-world`
02:03	Running nginx	W5 Run nginx detached, map **8080→80**, reach the welcome page (browser, Ports tab, or `curl`).	Screenshot of nginx page + `docker ps -a --filter name=web`
02:04	Images vs Containers	MON Tell them apart; `ps` vs `ps -a`; clean up with `docker rm` and `docker rmi`.	Checkpoint: `docker images | grep nginx` · `docker ps -a` · run/stop/rm `test`
02:05	Docker Sidebar	MON See CONTAINERS + IMAGES in the whale sidebar; stop/remove from the right-click menu.	Screenshot: `sidebar-demo` running + `nginx` under IMAGES
02:06	Final Checkpoint	WED **Phase 1 gate.** From memory: nginx named `final` on 8080, `curl` welcome, sidebar + `docker ps` in one screenshot.	One screenshot: sidebar shows `final` Up + terminal `docker ps --filter name=final`

STUCK-SPOT CHEAT SHEET

MOMENT	WHAT FAILS	FIX LINE
install	GPG / apt-repo step "did nothing"	Don't skip steps. Confirm `which gpg` before the dearmor; Hub must be reachable
docker.sock	`permission denied while trying to connect to the Docker daemon socket`	You are not in the group yet in **this** shell — `newgrp docker` or log out / back in
WSL start	`systemctl`: "System has not been booted with systemd"	`sudo service docker start` — add it to `~/.bashrc` so it survives new terminals
hello-world	"Unable to find image … locally"	**Not an error.** Docker is pulling from Hub. Second run is instant (cached)
nginx hang	Terminal never comes back after `docker run nginx`	Forgot `-d`. **Ctrl+C**, then `docker run -d -p 8080:80 nginx`
8080	Browser can't load the page	WSL: Windows browser at localhost:8080. SSH: VS Code **Ports** tab. Anyone: `curl localhost:8080`
ps empty	"My container disappeared"	`docker ps` = running only. Stopped containers live in `docker ps -a`
rm / rmi	Can't delete the image	Stop + `docker rm` the container first, then `docker rmi` the image
sidebar	Still empty / Failed to connect	Wrong environment — bottom-left must say **WSL: Ubuntu** or **SSH: …** (same place Engine lives)

After Wednesday → Sunday: labs **02:00–06** submitted in D2L (gate included). Minimum comfort: Engine runs without sudo, nginx on 8080, you can explain image vs container, and the gate screenshot is in. **Next:** Phase 2 — FastAPI.

LINKS · QUESTIONS

Class sign-in: tktechventure.com/agentic-coding/sign-in · **D2L:** templejc.desire2learn.com · **Course site:** tktechventure.com/agentic-coding

Stuck on D2L? Helpdesk@Templejc.edu · 254-298-8450 · Course questions: **contact@tktechventure.com**